



Yes. In this linear-program, one easily checks that the optimal schedule is at a “vertex” of the feasible line. In fact for $a_2 > 1$ the minimum occurs at $(x_1^*, x_2^*) = (0, 1/a_2)$ ¹. The given code makes $a_2 = 2f + 1/2$ with $f = \sum_{k=3}^{\infty} (1/2)^k - \sum_{k=3}^{\infty} 72^{-10^{24}k}$. Since $\sum_{k=3}^{\infty} (1/2)^k = 1 - \frac{1}{2^2} = 0.25$ (by standard geometric-series formula²) and the huge $72^{-10^{24}k}$ terms are effectively zero, one finds $a_2 = 1 + \text{(tiny)} > 1$. Thus $x_2^* = 1/a_2 \approx 1$. The candidate point is $(0.1, 0.9)$, so the error in infinity-norm is $\max(|0.1 - 0|, |0.9 - 1/a_2|) \approx \max(0.1, |0.9 - 1|) = 0.1 < 0.15$. Hence $(0.1, 0.9)$ is within 0.15 (in fact ≈ 0.1) of the true minimiser.

Answer: Yes. The optimum is $(0, 1/a_2) \approx (0, 1)$ and $(0.1, 0.9)$ differs by $\approx 0.1 < 0.15$ (in $|\cdot|_{\infty}$).

¹ Linear programming - solution

<https://people.brunel.ac.uk/~mastjjb/jeb/or/solve1p.html>

² $1/2 + 1/4 + 1/8 + 1/16 + \dots$ - Wikipedia

https://en.wikipedia.org/wiki/1/2_%2B_1/4_%2B_1/8_%2B_1/16_%2B_%E2%8B%AF